EE1101: Circuits and Network Analysis

Assignment - 10

Handed out: 25 - Oct - 2024 **Due : 04 - Nov - 2024 (before 5 PM)**

Instructions :

1. Please upload your assignment solutions to the course page on the Canvas platform. Only solutions submitted through canvas will be reviewed. For specific guidelines, refer to the instructions provided on the course page.
2. It is suggested that you attempt all the problems. However, it is sufficient to submit solutions for problems that total 10 points.
3. Submissions received after the deadline will attract negative marking. Ensure that your submissions are named in the following format: RollNo-Assignment-10.pdf.

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1. (3 points) Compute the admittance, hybrid and transmission parameters of the two port network shown in Fig. 1.

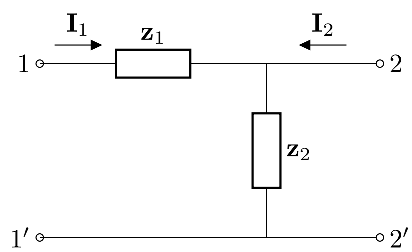


Fig. 1: Two port network for question 1

2. (2 points) Compute the hybrid and admittance parameters of the two port network shown in Fig. 2.

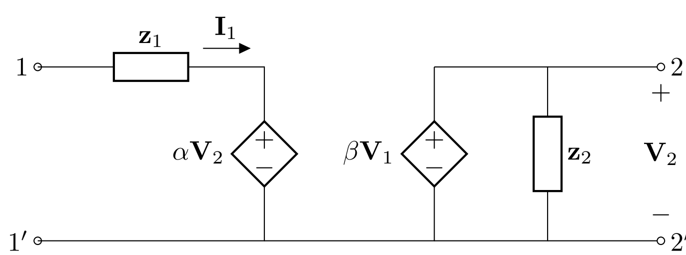


Fig. 2: Two port network for question 2

3. (3 points) For the circuit shown in Fig. 3, compute the Thevenin equivalent circuit as seen from the port *a*-*o*. The two-port network is characterized by the admittance parameters $\mathbf{y}_{11}$, $\mathbf{y}_{12}$, $\mathbf{y}_{21}$ and $\mathbf{y}_{22}$.
4. (8 points) Consider the circuit shown in Fig. 1 with $\mathbf{z}_1 = j\omega L$ and $\mathbf{z}_2 = \frac{1}{j\omega C}$.

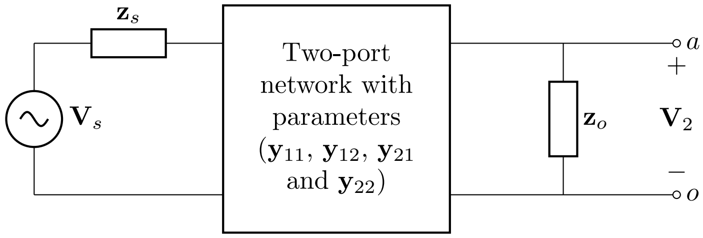


Fig. 3: Circuit for Question 3

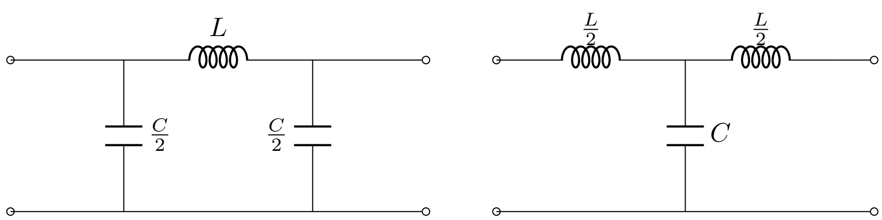
(a) (2 points) Compute the equivalent transmission parameters of a two port network obtained by when two such two-port networks are cascaded.

(b) (2 points) Compute the equivalent transmission parameters of a two port network obtained by connecting two such two-port networks are connected in series.

(c) (2 points) Compute the equivalent transmission parameters of a two port network obtained when three such two-port networks are cascaded.

(d) (2 points) Compute the equivalent transmission parameters of a two port network obtained when N such two-port networks are cascaded.

5. (4 points) Compute the impedance and admittance parameters of the two port networks shown in Fig. 4.



(a) Network 1 (b) Network 2

Fig. 4: Two port networks for Questions 5-7

6. (2 points) Compute the equivalent transmission parameters when the two port networks shown in Fig. 4 are connected in cascade manner. (assume the cascade connection to be such that the output of network 1 acts as an input to network 2).

7. (2 points) Compute the equivalent hybrid parameters when the two port networks shown in Fig. 4 are connected in series.